Nodal DG method for the 1D advection equation

This text was mainly inspired by [1]. My own code supporting this text can be found on Github.

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Symbols and definitions

- u_h - approximate solution obtained on a mesh with step h
- $(u, v)_{L^2(\Omega)} = \int_\Omega uv \, \mathrm{d}x$ - scalar product in L^2
- $\| u \|^2_{L^2(\Omega)} = (u, u)_{L^2(\Omega)}$
- $P_j\big(D^k\big)$ - space of polynomoials of up to order j on an interval D^k
- φ_i, ψ_i - i-th basis function
- u^- - limit of u when approaching the boundary of an element from the left
- u^+ - limit of u when approaching the boundary of an element from the right
- $f^* = f^*(u^-, u^+)$ - numerical flux
- $\hat{\boldsymbol{n}}$ - outward unit normal
- $\{\!\{u\}\!\} = \frac{u^- + u^+}{2}$
- $[\![u]\!] = \hat{\boldsymbol{n}}^- u^- + \hat{\boldsymbol{n}}^+ u^+$
- $[\![\boldsymbol{u}]\!] = \hat{\boldsymbol{n}}^- \cdot \boldsymbol{u}^- + \hat{\boldsymbol{n}}^+ \cdot \boldsymbol{u}^+$
- $l_i(x)$ - i-th Lagrange polynomoial
- N - order of polynomial approximation
- $N_p = N + 1$ - number of points in each element (number of degrees of freedom)

1. Problem overview

The general one dimensional advection equation is of the following form:

$$\frac{\partial \boldsymbol{u}}{\partial t} + \frac{\partial \boldsymbol{f}(\boldsymbol{u})}{\partial x} = \boldsymbol{s}(x,t) \quad [x,t] \in \langle L, R\rangle \times \mathbb{R}^+$$

where $\boldsymbol{f} = [f_1(\boldsymbol{u}), f_2(\boldsymbol{u}), ..., f_n(\boldsymbol{u})]^T$ is the physical flux, $\boldsymbol{u} = \boldsymbol{u}(x,t) = [u_1, u_2, ..., u_n]^T$ is the unknown vector valued function and $\boldsymbol{s}(x,t)$ is the source term. This equation is hyperbolic.

For this problem to be well defined we need to impose initial and boundary conditions.

$$\begin{aligned} \boldsymbol{u}(x,0) &= \boldsymbol{u}_0(x) \\ \mathcal{B}_L \boldsymbol{u}(L,t) &= \boldsymbol{g}_1(t) \quad \text{at} \quad x = L \\ \mathcal{B}_R \boldsymbol{u}(R,t) &= \boldsymbol{g}_2(t) \quad \text{at} \quad x = R \end{aligned}$$

where the sum of the ranks of the boundary operators $\mathcal{B}_L, \mathcal{B}_R$ equals the number of the required inflow conditions.

1.1. Weak formulation of the problem

To obtain the weak formulation we multiply the equation by a test function $\boldsymbol{v} \in V$ and integrate over the domain Ω.

$$\int_\Omega \boldsymbol{v}^T \frac{\partial \boldsymbol{u}}{\partial t}\,\mathrm{d}x + \int_\Omega \boldsymbol{v}^T \frac{\partial \boldsymbol{f}(\boldsymbol{u})}{\partial x}\,\mathrm{d}x = \int_\Omega \boldsymbol{v}^T \boldsymbol{s}(x,t)\,\mathrm{d}x.$$

Using per-partes on the second integral we obtain

$$\int_\Omega \boldsymbol{v}^T \frac{\partial \boldsymbol{u}}{\partial t}\,\mathrm{d}x - \int_\Omega \frac{\partial \boldsymbol{v}^T}{\partial x} \boldsymbol{f}(\boldsymbol{u})\,\mathrm{d}x + \int_{\partial\Omega} \hat{\boldsymbol{n}} \cdot \boldsymbol{v}^T \boldsymbol{f}(\boldsymbol{u})\,\mathrm{d}x = \int_\Omega \boldsymbol{v}^T \boldsymbol{s}(x,t)\,\mathrm{d}x,$$

$\hat{\boldsymbol{n}}$ here is a unit outward normal. Notice that we express the term $\left[\boldsymbol{v}^T \boldsymbol{f}(\boldsymbol{u})\right]_L^R$ in a integral form $\int_{\partial\Omega} \hat{\boldsymbol{n}} \cdot \boldsymbol{v}^T \boldsymbol{f}(\boldsymbol{u})\,\mathrm{d}x$, this will later help us when generalazing the scheme.

2. Discontinous Galerkin Method

Let's focus on linear scalar advection equation for now and generalize the scheme later.

2.1. Spatial discretization

We split our domain ($\Omega = \langle L, R\rangle$) into N elements $D^j = \langle x_{j-\frac{1}{2}}, x_{j+\frac{1}{2}}\rangle$, $j = 1, 2, ..., N$, here x_j is the center of the element and $x_{\frac{1}{2}} = L$, $x_{N+\frac{1}{2}} = R$.

$$\Omega \approx \Omega_h = \bigcup_{k=1}^{K} D^k$$

We now formulate the local weak formulation

$$\int_{D^k} v \frac{\partial u}{\partial t}\,\mathrm{d}x = \int_{D^k} \frac{\partial v}{\partial x} f(u)\,\mathrm{d}x - \int_{\partial D^k} \hat{\boldsymbol{n}} v f(u)\,\mathrm{d}x$$

But now we have a problem, because u is double valued at the boundaries (of each element). To solve this we define a numerical flux $f^* = f^*(u^-, u^+)$. The equation then becomes

$$\int_{D^k} v \frac{\partial u}{\partial t}\,\mathrm{d}x = \int_{D^k} \frac{\partial v}{\partial x} f(u)\,\mathrm{d}x - \int_{\partial D^k} \hat{\boldsymbol{n}} v f^*(u)\,\mathrm{d}x$$

The flux must be consistent i.e. $f^*(a,a) = f(a)$. One such numerical flux could be the local Lax-Friedrichs numerical flux.

$$f^* = \{\!\{f(u)\}\!\} + \frac{C}{2}[\![u]\!]$$

where the local constant C is determined by the maximum eigenvalue (the spectral radius) of the physical flux Jacobi matrix.

$$C = \max_i \lambda_i = \rho(f_u) = \rho\left(\frac{\partial f}{\partial u}\right)$$

Or the upwind flux

...

2.2. Basis functions

We assume that the approximate solution $\boldsymbol{u}_h(x,t)$ can be expressed as a direct sum of local piecewise polynomial solutions

$$u(x,t) \approx u_h(x,t) = \bigoplus_{k=1}^{K} u_h^k\big(x^k, t\big)$$

We define a local function space V_h^k such that

$$V_h = \bigoplus_{k=1}^{K} V_h^k$$

$$V_h^k = \big\{\varphi \in L^2\big(D^k\big) : \varphi\,|_{D^k} \in P_j\big(D^k\big),\, j = 1, 2, ..., N\big\},$$

where $P_j\big(D^k\big)$ is a space of polynomials of at most degree k on the interval D^j.

Now we can express the local approximate solution in the basis of the space V_h^k

$$x \in D^k \quad : \quad u_h^k = \sum_n^{N_p} \hat{u}_n^k(t)\varphi_n(x) = \sum_i^{N_p} u_h^k(x_i, t) l_i(x),$$

where $\hat{\boldsymbol{u}}_n$ is a vector of coefficients and l_i is the i-th interpolating Lagrange polynomial and $x_i \in D^k$ are distinct. The first expression is said to be modal representation and the second nodal. We won't discuss the modal formulation in this text.

Following the Galerkin approach we replace the test function with each of the basis functions. This yields a system of $N+1$ equations for each element

$$\int_{D^k} l_i(x) \frac{\partial}{\partial t}\left(\sum_{j=1}^{N_p} u_h\big(x_j, t\big)\right) \mathrm{d}x = \int_{D^k} \frac{\mathrm{d}l_i(x)}{\mathrm{d}x} \sum_{j=1}^{N_p} a u_h\big(x_j, t\big) l_j(x)\,\mathrm{d}x - \int_{\partial D^k} l_i f^*\,\mathrm{d}x$$

$$\frac{\mathrm{d}}{\mathrm{d}t} \sum_{j=1}^{N_p} u_h\big(x_j, t\big) \int_{D^k} l_i(x) l_j(x)\,\mathrm{d}x = \sum_{j=1}^{N_p} a u_h\big(x_j, t\big) \int_{D^k} \frac{\mathrm{d}l_i(x)}{\mathrm{d}x} l_j(x)\,\mathrm{d}x - \int_{\partial D^k} l_i f^*\,\mathrm{d}x$$

$$i = 1, 2, ..., N+1$$

We can rewrite this equation into it's matrix form

$$M^k \frac{\mathrm{d}\boldsymbol{u}_h^k}{\mathrm{d}t} = a\big(S^k\big)^T \boldsymbol{u}_h^k - \mathcal{E}\boldsymbol{f}_*^k,$$

where

$$M_{ij}^k = \int_{D^k} l_i(x) l_j(x)\,\mathrm{d}x \quad \text{- the mass matrix}$$

$$S_{ij}^k = \int_{D^k} l_i(x) \frac{\mathrm{d}l_j(x)}{\mathrm{d}x}\,\mathrm{d}x \quad \text{- the stifness matrix}$$

$$\mathcal{E} = \begin{cases} -1 & : \text{ lower right corner} \\ 1 & : \text{ upper left corner} \\ 0 & : \text{ otherwise} \end{cases}$$

$$(\boldsymbol{u}_h)_j^k = u_h\left(x_j^k, t\right) \quad \text{- the vector of unknowns}$$

$$(\boldsymbol{f}_h)_j^k \quad \text{- the physical flux vector}$$

$$\boldsymbol{f}_*^k = \left(f^*\left(x_L^k\right), f^*\left(x_R^k\right)\right)^T \quad \text{- numerical flux at endpoints}$$

This matrix equation is the semi-discrete form of the PDE. We will discuss all the local operatros in more detail later.

When choosing the basis of the space V_h^k we have many options. But for the mass matrix to be well conditioned we will choose the Legendre polynomials.

2.2.1. Legender polynomials

Legendre polynomials are a complete set of orthogonal polynomials defined on $\langle -1, 1\rangle$

An easy way to calculate the polynomials is through the three term reccurence using the Bonnet's formula [cite]

$$P_n = \frac{(2n+1)xP_{n-1}(x) - nP_{n-2}(x)}{n+1};$$

To start the reccurence we need the first two terms. They are given as

$$P_0(x) = 1, \quad P_1(x) = x$$

Now to obtain the normalized (in the L^2 norm) Legender polynomials $\tilde{P}_n(x)$ we multiply each polynomial by an appropriate coefficient

$$\tilde{P}_n(x) = \sqrt{\frac{2n+1}{2}} P_n(x)$$

To obtain a derivative of the Legendre polynomial we use the formula [cite]

$$P_n'(x) = \frac{n(P_{n-1}(x) - xP_n(x))}{1-x^2}$$

Notice that the formula doesn't hold for $x = \pm 1$, for that we use a the following property

$$P_n'(1) = \frac{n(n+1)}{2}$$

and

$$P_n'(-x) = (-1)^{2n-1} P_n'(x)$$

Important properties

1. $$P_n(-x) = (-1)^n P_n(x)$$
2. $$\int_{-1}^{1} P_n(x)P_m(x)\,\mathrm{d}x = 0 \quad \text{for} \quad n \neq m$$
3. $$\int_{-1}^{1} P_n(x)\,\mathrm{d}x = 0 \quad \text{for} \quad n \geq 1$$

2.3. Nodal representation

Let's now discuss the nodal representation in greater detail. We define $\hat{\boldsymbol{u}}_n$ such that the approximation is interpolatory i.e.

$$u(x_i) = \sum \hat{u}_n \psi_n(x_i),$$

where x_i represents distinct grid nodes. We can now write

$$\boldsymbol{u} = \mathcal{V}\hat{\boldsymbol{u}},$$

where $\mathcal{V}_{ij} = \psi_j(x_i)$ is the Vandermonde matrix and $u_i = u(x_i)$. This matrix connects the modes $\hat{u}$ and nodal values $u(x_i)$. We would like the matrix to be well conditioned. We already chose the basis polynomials, so only the nodes x_i are left, to define the Vandermonde matrix. The node's that acomplish this are the solution of

$$(1 - x^2)\frac{\mathrm{d}}{\mathrm{d}x}P_N(x) = 0$$

these are known as the Legendre-Gauss-Lobatto nodes. The LGL nodes will be noted by the greek letter ξ. There are $N + 1$ LGL nodes for an approximation of order N.

2.4. Local operators

Up till now we were developing a sensible local representation of the approximate solution. We should now discuss the various local operators in the nodal formulation of DGM.

2.4.1. Mass matrix

The local mass matrix M^k is given as

$$M^k_{ij} = \int_{x^k_l}^{x^k_r} l^k_i(x) l^k_j(x)\,\mathrm{d}x = \frac{h_k}{2}\int_{-1}^{1} l_i(r) l_j(r)\,\mathrm{d}r = \frac{h_k}{2}\big(l_i, l_j\big)_{L^2} = \frac{h_k}{2}M_{ij},$$

where the coefficient $\frac{h_k}{2}$ is a Jacobian coming from the affine transformation and M is the mass matrix defined on the reference element.

We know that

$$l_i(r) = \left(\mathcal{V}^T\right)^{-1}_{in}\varphi_n(r).$$

From which we get

$$M_{ij} = \int_{-1}^{1} \left(\mathcal{V}^T\right)^{-1}_{in}\varphi_n(r)\left(\mathcal{V}^T\right)^{-1}_{jm}\varphi_m(r)\,\mathrm{d}r = \left(\mathcal{V}^T\right)^{-1}_{im}\left(\mathcal{V}^T\right)^{-1}_{jm}(\varphi_n, \varphi_m)_{L^2} =$$
$$= \left(\mathcal{V}^T\right)^{-1}_{im}\left(\mathcal{V}^T\right)^{-1}_{jm}$$

Thus

$$M^k = \frac{h_k}{2} M = \frac{h_k}{2} (\mathcal{V}\mathcal{V}^T)^{-1}$$

2.4.2. Stiffness matrix

The local stiffness matrix is given as

$$S_{ij}^k = \int_{x_l^k}^{x_r^k} l_i^k(x) \frac{\mathrm{d}}{\mathrm{d}x} l_j^k(x) \,\mathrm{d}x = \int_{-1}^{1} l_i(r) \frac{\mathrm{d}}{\mathrm{d}r} l_j(r) \,\mathrm{d}r = S_{ij}$$

To simply computate this we define the differentiation matrix.

$$D_{ij} = \frac{\mathrm{d}}{\mathrm{d}r} l_j \mid_{r_i}$$

Consider the product

$$(MD)_{ij} = M_{in} D_{nj} = \sum_n \int_{-1}^{1} l_i(r) l_n(r) \frac{\mathrm{d}}{\mathrm{d}r} l_j \mid_{r_n} = \int_{-1}^{1} l_i(r) \sum_n \frac{\mathrm{d}l_j}{\mathrm{d}r} \mid_{r_n} l_n(r) \,\mathrm{d}r =$$

$$= \int_{-1}^{1} l_i(r) \frac{\mathrm{d}l_j}{\mathrm{d}r} \,\mathrm{d}r = S_{ij}$$

$$MD = S$$

The entries of the differentiation matrix can be found directly

$$D = \mathcal{V}_r \mathcal{V}^{-1},$$

where $\mathcal{V}_r$ is the Vandermonde matrix assembled from differentiated Legendre polynomials.

$$(\mathcal{V}_r)_{ij} = \psi_j'(\xi_i)$$

But for the weak formulation we need S^T, for that we use the identity

$$S^T = D^T M$$

To recover the semi-discrete formulation we need to multiply this by the mass matrix from the left

$$M^{-1} S^T = M^{-1} D^T M = \mathcal{V}\mathcal{V}_r^T M = D_w$$

We call this matrix D_w, differentiation matrix for the weak form.

2.4.3. Surface integral

This operator is responsible of extracting the surface terms of the form

$$\int_{\partial D^k} \hat{\boldsymbol{n}} f^* l_i(r) \,\mathrm{d}x = f^* \mid_{x_R} \boldsymbol{e}_n - u^* \mid_{x_L} \boldsymbol{e}_1,$$

where $\boldsymbol{e}_i$ is a zero vector with 1 at index i. We can rewrite this using a $N_p \times 2$ matrix $\mathcal{E}$ that is zero except for upper right corener that is 1 and the lower left corner that is -1 like so

$$\int_{\partial D^k} \hat{\boldsymbol{n}} f^* l_i(r) \,\mathrm{d}x = \mathcal{E} \cdot (f_R^*, f_L^*)^T$$

Again, to recover the semi-discrete form we multiply by the inverse mass matrix from the left. We call this matrix the LIFT.

$$\text{LIFT} = M^{-1}\mathcal{E} = \mathcal{V}\mathcal{V}^T\mathcal{E}$$

2.5. Time discretization

Two explicit RK4 methods were implemented to integrate the semidiscrete system

$$\frac{\mathrm{d}u_h}{\mathrm{d}t} = \mathcal{L}_h(u_h, t)$$

2.5.1. ERK

General explicit four stage Runge-Kutta method.

$$\begin{aligned}
k_1 &= \mathcal{L}_h(u_h^n, t^n) \\
k_2 &= \mathcal{L}_h\left(u_h^n + \frac{1}{2}\Delta t k_1, t^n + \Delta t\right) \\
k_3 &= \mathcal{L}_h\left(u_h^n + \frac{1}{2}\Delta t k_2, t^n + \Delta t\right) \\
k_4 &= \mathcal{L}_h\left(u_h^n + \frac{1}{2}\Delta t k_3, t^n + \Delta t\right) \\
u_h^{n+1} &= u_h^n + \frac{1}{6}\Delta t(k_1 + k_2 + k_3 + k_4)
\end{aligned}$$

2.5.2. LSERK

Low storage explicit five stage Runge-Kutta method.

2.6. Example linear problem

We are given the following problem

$$\partial_t u + \partial_x u = 0, \quad x \in \langle 0, 2\pi \rangle$$

with periodic boundary conditions.

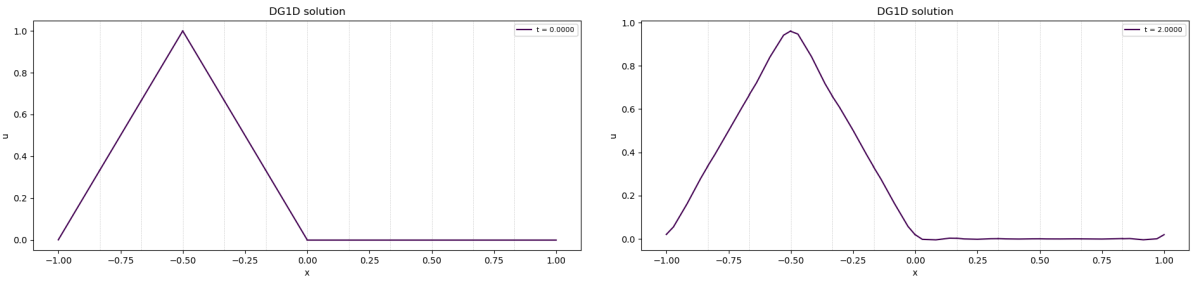
We assume the solution can be approximated as

$$u_h^k = \sum_i u_h^k(x_i^k, t) l_i^k(x).$$

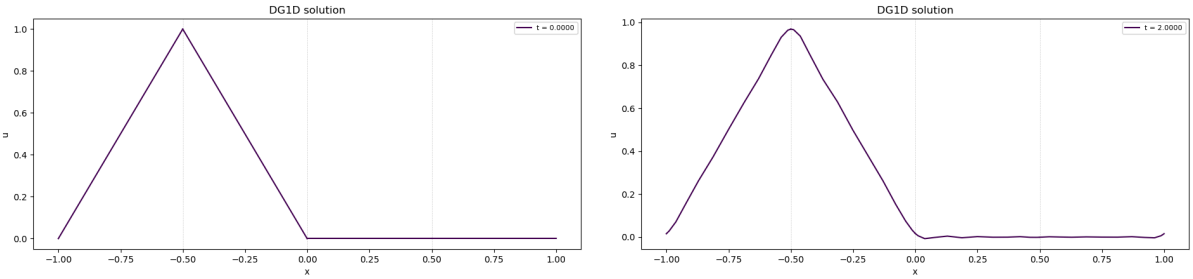
And that the direct sum of these solutions is an approximation to the global solution. This yields the local semidiscrete scheme

$$\begin{aligned}
M^k \frac{\mathrm{d}u_h^k}{\mathrm{d}t} - S^T u_h^k &= -[l^k(x)(u)^*)]_{x_l^k}^{x_r^k} = -\mathcal{E}((u)^*, (u)^*)^T \\
J^k M \frac{\mathrm{d}u_h^k}{\mathrm{d}t} - S^T u_h^k &= -\mathcal{E}((u)^*, (u)^*)^T \\
\frac{\mathrm{d}u_h^k}{\mathrm{d}t} &= \frac{1}{J^k}\left(M^{-1}S^T u_h^k - M^{-1}\mathcal{E}(f_R^*, f_L^*)^T\right) \\
\frac{\mathrm{d}u_h^k}{\mathrm{d}t} &= \frac{1}{J^k}\left(D_w u_h^k - \text{LIFT } (f_R^*, f_L^*)^T\right)
\end{aligned}$$

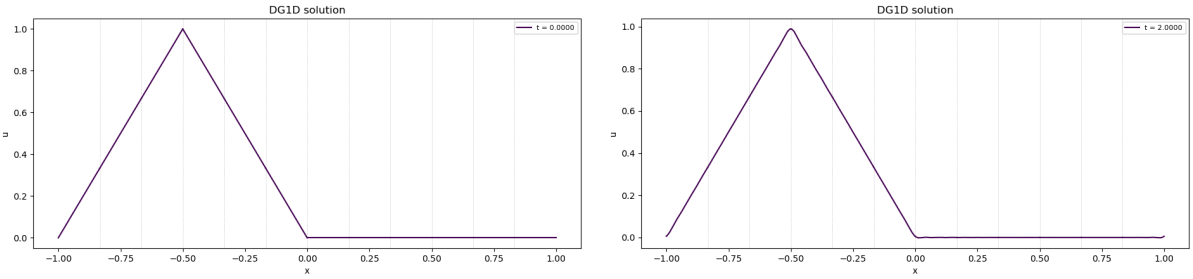
Now let's try multiple initial conditions with different number of elemnets and polynomial order. First we will test non-smooth initial conditions.



Cone with $K = 12$, $N = 4$

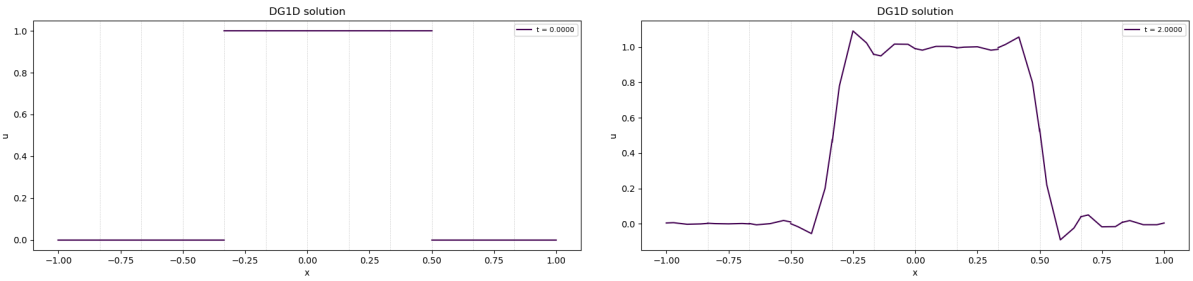


Cone with $K = 4$, $N = 12$

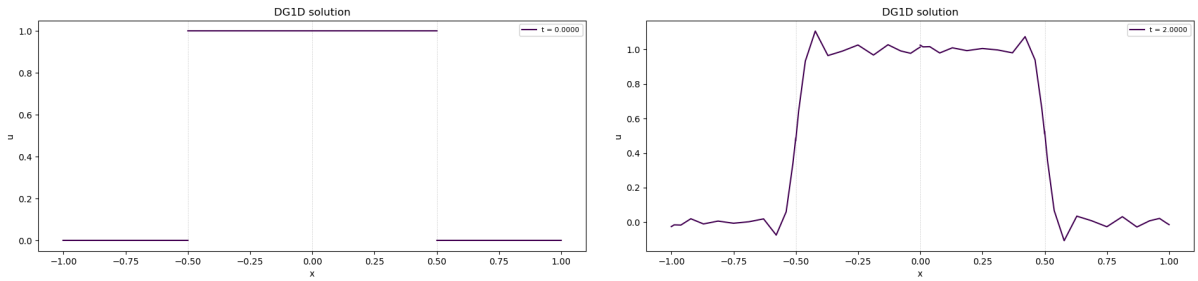


Cone with $K = 12$, $N = 12$

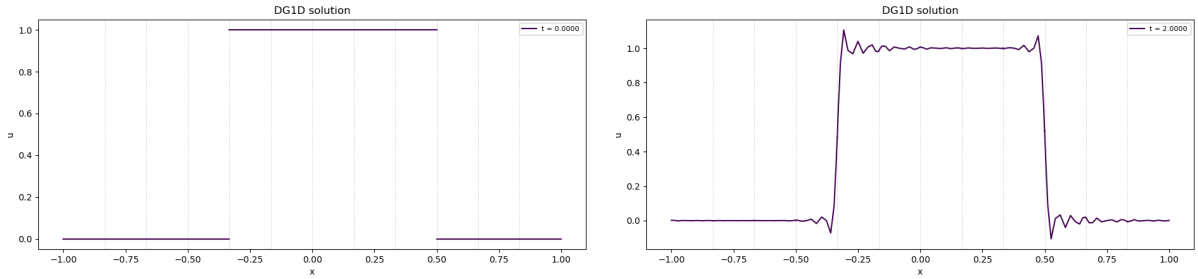
Now let us test a non-continuous initial condition.



Step signal with $K = 12$, $N = 4$



Step signal with $K = 4$, $N = 12$



Step signal with $K = 12$, $N = 12$

3. Nonlinearity

3.1. Filters

3.2. Limiters

3.3. Example nonlinear problem

burger's equation

4. Systems of equations

4.1. Euler's equation

4.1.1. Sod's problem

Bibliography

[1] J. S. Hesthaven and T. Warburton, *Nodal discontinuous Galerkin methods: algorithms, analysis, and applications*. Springer, 2008.